Why Genome Selection Matters in Molecular Diagnostic Assay Design

Lessons from genome provenance screening in *Chlamydia trachomatis* and the Swedish plasmid variant

THE MAIN IDEA

Reference-genome selection is part of diagnostic assay design. Verify the population before evaluating conservation—and consider variation beyond the reference set.

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Executive summary

The choice of reference genomes influences how target conservation is assessed in molecular diagnostic assay development. Public databases contain valuable sequence data, but not every record is suitable for estimating variation across the population an assay is intended to detect. Genome provenance, assembly quality and biological context therefore need to be checked before inclusivity analysis begins.

A computational target-selection project for *Chlamydia trachomatis* illustrates this approach. The genome inventory included 125 complete RefSeq assemblies and 149 additional GenBank-only genomes. Screening the records identified the 149 GenBank-only genomes as engineered laboratory strains, and they were excluded from the reference set. Further checks of the RefSeq collection separated 97 primary genomes from 10 requiring quality review and 18 excluded records. The resulting primary set was used for pangenome analysis and target consensus construction.

Target choice also requires attention to variation that may not be represented in the selected genomes. The Swedish *C. trachomatis* plasmid variant, which was missed by some commercial diagnostic tests, is a useful example. It supports considering a second, independent genomic target when designing an assay. This paper focuses on genome selection and that biological rationale; the proposed chromosomal-plus-plasmid duplex remains a computational design requiring laboratory evaluation.

1. Selecting an appropriate reference genome set

The project aimed to identify suitable targets for a proposed *C. trachomatis* duplex qPCR/TaqMan assay, combining a chromosomal locus with a target on the multicopy cryptic plasmid. Before assessing conservation, the available genome records were reviewed to determine whether they represented the intended diagnostic population.

The initial inventory comprised 125 complete RefSeq assemblies [2] and 149 non-RefSeq (GenBank-only) genomes. A provenance check traced all 149 additional records to a single BioProject describing deliberately engineered laboratory strains, including transposon mutants and antibiotic-resistance-tagged constructs. These records were excluded from the reference set used for conservation analysis. A gene deliberately disrupted in the laboratory could otherwise be interpreted as naturally absent from some strains.

The issue was not the technical quality of those engineered sequences. They may be entirely suitable for the studies that generated them, but they were not appropriate for estimating conservation across natural isolates. An accurate calculation based on an unsuitable genome collection can still lead to a misleading assessment of a diagnostic target.

2. What the RefSeq genome audit found

The 125 complete RefSeq assemblies were then checked individually rather than accepted solely because of their database status. Checks covered accession and file integrity, genome provenance, assembly quality, sequence-level species consistency and agreement between serovar labels and *ompA* sequence relationships. Pairwise average nucleotide identity was at least 98.90% across the set, supporting species-level consistency, but this result alone could not distinguish natural isolates from engineered strains [1,3].

The audit divided the 125 assemblies into three groups:

97	10	18
Used for pangenome and target consensus analysis	Reviewed separately against the chromosomal oligonucleotide sets	Excluded from the primary reference set

Eighteen RefSeq assemblies had engineering or mutagenesis indicators in their submission records and were excluded. Ten other genomes had concerns requiring separate review, including assembly-quality flags, documented laboratory passage or inconsistent metadata. The audit also had to account for experiments described in BioProject titles rather than strain names alone. Among 104 genomes with serovar labels, three labels differed from their nearest *ompA* sequence neighbours; these cases were recorded for review rather than automatically treated as correct or incorrect [1].

Only the 97 primary genomes contributed to pangenome analysis and target consensus construction. The ten quality-review genomes were retained for a separate check: each was compared with each of the seven chromosomal oligonucleotide sets, giving 70 genome-by-set comparisons (10 × 7). Of these, 69 matched exactly and one carried a single probe mismatch with both primers exact; none was classified as at risk or failed. This provided additional evidence about candidate inclusivity without giving flagged genomes equal weight in defining the primary reference set [1].

Together, these checks established which genomes could support the conservation analysis, which needed further scrutiny and which should not be used for that purpose. The relevant question was not simply whether a sequence had a RefSeq accession, but whether it was appropriate evidence for the intended assay.

3. Why plasmid variation matters for target selection

Selecting suitable genomes does not remove every biological risk. A target region can be conserved in the available reference set yet vary in strains not represented by that set. The *C. trachomatis* cryptic plasmid illustrates this limitation. Its multiple copies make it attractive for molecular detection, but changes in a targeted region can compromise plasmid-only assays.

In 2006, a *C. trachomatis* variant carrying a 377-base-pair deletion in the cryptic plasmid was identified in Sweden. The deletion of regions targeted by certain commercial tests led to false-negative results. This was a failure of particular assay targets, not evidence that plasmid-based detection is inherently unreliable [4,5].

For the present project, a duplex assay was proposed with one conserved chromosomal target and one plasmid target. The plasmid may provide a sensitivity advantage through its copy number, while the chromosomal locus offers a separate genomic target if a plasmid region changes. This architecture is a reasonable design choice, but it does not demonstrate that the selected oligonucleotides detect every plasmid variant. Sensitivity, inclusivity and duplex compatibility remain to be tested experimentally.

An important uncertainty was documented rather than resolved by assumption. Although the Swedish 377-bp deletion is well established, its specific coordinates could not be independently confirmed against the primary sources available during this project. No claim was therefore made that the selected plasmid oligonucleotides avoided the deletion on the basis of those coordinates. The variant informed the decision to include an independent chromosomal target; it did not provide direct validation of the proposed assay [1,4].

4. Practical implications for assay development

These examples concern two separate decisions. The genome audit established a suitable population for assessing target conservation. The plasmid example shows why conservation within that population is not, by itself, enough to establish the robustness of a diagnostic target. Both decisions belong early in assay development, before detailed primer and probe screening.

A defensible approach is to specify the intended diagnostic population, check the provenance and quality of reference genomes, document which records contribute to inclusivity calculations and examine flagged genomes separately where appropriate. Known biological variation should also inform whether a second genomic target could reduce reliance on a single locus. The subsequent oligonucleotide-design workflow can then be built on clearer biological assumptions.

Conclusion

In this *C. trachomatis* project, genome screening defined the reference set used to assess target conservation. It identified 149 engineered GenBank-only records and additional RefSeq assemblies that were unsuitable for the primary reference population. The resulting 97-genome set provided a more clearly defined basis for target selection, supported by separate assessment of ten quality-review genomes.

The Swedish plasmid variant illustrates a different consideration: even a carefully selected reference population cannot account for every possible change in a diagnostic target. Combining a conserved chromosomal locus with a plasmid target was proposed to avoid relying on the plasmid alone. The design still needs experimental evaluation. These findings highlight the importance of evaluating genome provenance, sequence quality and biological variation before selecting diagnostic targets. Reference-genome selection should be treated as an integral part of assay development, rather than a routine data-preparation step.

References and supporting resources

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